OVER A BASE CARRYING AN AMPLE INVERTIBLE SHEAF, PROJECTIVE MORPHISMS ARE H-PROJECTIVE AND QUASI-PROJECTIVE ONES ARE H-QUASI-PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-projective-over-quasi-projective-is-H-projective is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 10847-10854 and proof span 10856-10882 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof uses the author's identification of a projective bundle of a free module with a projective space over the base and the invariance of projective bundles under twisting; both are preserved as written.

Editorial note. 6 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A scheme carrying an ample invertible sheaf is quasi-compact and separated, and it admits an immersion into a projective space over the integers, cited from the properties and constructions chapters. Cited at proof lines 10856-10860. Supplies the hypotheses under which the passage from H-projective to H-quasi-projective is available.
- The statement that over a base with those properties a quasi-projective morphism is obtained from an H-projective one, cited from the same chapter. Cited at proof lines 10859-10861. Gives that the first assertion implies the second.
- The construction of a morphism with a projective bundle as source from a generating set of sections of the corresponding module, the identification of a projective bundle of a free module with a projective space over the base, and the invariance of a projective bundle under twisting the module by an invertible sheaf. Cited at proof lines 10868-10881. Produces the closed immersion into a projective space over the base in the generated case, and reduces the general case to it by an ample twist.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01MC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2117–2117. Same-chapter reference used in the retained author text at source line(s) 9142, 10597, 10860. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02NC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3475–3475. Same-chapter reference used in the retained author text at source line(s) 10880. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 07RM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9872–9872. Same-chapter reference used in the retained author text at source line(s) 10508, 10792, 10861. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original definition 01PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2241–2241. Same-chapter reference used in the retained author text at source line(s) 8623, 8846, 10858. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01Q1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2250–2250. Same-chapter reference used in the retained author text at source line(s) 8742, 10859, 10909. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original proposition 01Q3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2252–2252. Same-chapter reference used in the retained author text at source line(s) 8676, 8876, 9431, 9484, 9655, 10874. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme which admits an ample invertible sheaf. Then*

(1) *any projective morphism $X \to S$ is H-projective, and*
(2) *any quasi-projective morphism $X \to S$ is H-quasi-projective.*

Proof. The assumptions on S imply that S is quasi-compact and separated, see Properties, Definition 01PS and Lemma 01Q1 and Constructions, Lemma 01MC. Hence Lemma 07RM applies and we see that (1) implies (2). Let $\mathcal{E}$ be a finite type quasi-coherent $\mathcal{O}_S$-module. By our definition of projective morphisms it suffices to show that $\mathbf{P}(\mathcal{E}) \to S$ is H-projective. If $\mathcal{E}$ is generated by finitely many global sections, then the corresponding surjection $\mathcal{O}_S^{\oplus n} \to \mathcal{E}$ induces a closed immersion

$$\mathbf{P}(\mathcal{E}) \longrightarrow \mathbf{P}(\mathcal{O}_S^{\oplus n}) = \mathbf{P}_S^{n-1}$$

as desired. In general, let $\mathcal{L}$ be an ample invertible sheaf on S. By Properties, Proposition 01Q3 there exists an integer n such that $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{L}^{\otimes n}$ is globally generated

by finitely many sections. Since $\mathbf{P}(\mathcal{E}) = \mathbf{P}(\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{L}^{\otimes n})$ by Constructions, Lemma 02NC this finishes the proof. □